Safety-Critical Evasion for Quadrotor UAVs: NMPC with Embedded High-Order Control Barrier Functions Against an Adversarial Pursuer

Bryan S. Guevara, Luis F. Simancas, and Tiago Nascimento

***Abstract*—This paper addresses the one-vs-one aerial pursuit-evasion problem for quadrotor UAVs operating in a *bounded workspace*. We propose a Nonlinear Model Predictive Control (NMPC) framework for the evader in which two safety constraints are enforced simultaneously as *hard-constraint degree-2 High-Order Control Barrier Functions* (HOCBFs) embedded inside the Optimal Control Problem (OCP): (i) a minimum-separation barrier preventing capture, and (ii) a workspace-containment barrier bounding the flight volume. The two barriers have competing objectives near the workspace boundary under aggressive pursuit—the *cornering* scenario that makes post-hoc CBF-QP filters infeasible or generates chattering—and the receding-horizon optimizer resolves them jointly without explicit priority rules. The pursuer is modeled as an augmented constant-velocity state in the OCP, providing multi-step foresight over both barrier boundaries. Both barriers are shown to have relative degree 2 from the collective thrust T, yielding affine HOCBF conditions encoded as nonlinear stage constraints in acados with SQP-RTI. Both agents run independent solvers at $100\,\mathrm{Hz}$ (decoupled game architecture). Software-in-the-Loop experiments in MuJoCo with two full quadrotor models validate: (a) zero constraint violations including deliberate cornering scenarios where the pursuer drives the evader against the workspace boundary; (b) superiority over a post-hoc CBF-QP filter, which accumulates safety violations and physical collisions under the same conditions; and (c) real-time feasibility within a $3\,\mathrm{ms}$ budget at $100\,\mathrm{Hz}$.**

***Index Terms*—Pursuit-Evasion, Control Barrier Functions, Nonlinear Model Predictive Control, Quadrotor UAV, Adversarial Games, Workspace Constraints, Safety-Critical Control, acados.**

I. Introduction

ADVERSARIAL scenarios involving autonomous aerial vehicles have attracted increasing attention as unmanned aerial systems (UAS) are deployed in competitive or contested environments [19], [20]. The one-vs-one aerial pursuit-evasion game—where a pursuer attempts to minimize its distance to an evader while the evader strives to maximize it—is a canonical instance of a zero-sum differential game [25] that captures the essential tension between aggressive maneuvering and safety-critical separation guarantees. Beyond academic interest, the problem is directly relevant to counter-drone systems, autonomous air combat, and multi-agent coordination under adversarial conditions [26].

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A. Existing Approaches and Their Limitations

1) Reinforcement Learning Dominance: The dominant paradigm in recent drone pursuit-evasion literature is multi-agent reinforcement learning (MARL). Wang *et al.* [1] proposed an online planning framework using deep RL for multi-UAV pursuit-evasion, demonstrating emergent cooperative strategies. Subsequent works have advanced this direction with asynchronous multi-stage adversarial training [4], population-based self-play and adaptive teaming in real CrazyFlie deployments [3], and zero-shot sim-to-real transfer on three quadrotors with body-rate-level commands [2]. The 2025 work by Kaufmann *et al.* [5] further demonstrates that body-rate control enables significantly more agile evasion compared to velocity-level commands.

Despite these advances, RL-based approaches share a fundamental limitation: they provide no *formal guarantees* on constraint satisfaction. A learned policy may violate minimum separation requirements during exploration, generalize poorly to unseen adversary behaviors, and offers no certificate of forward invariance of the safe set. This gap is particularly critical in safety-certified airspace or in scenarios where physical contact implies mission failure.

2) Model Predictive Control Approaches: Model Predictive Control (MPC) provides a principled alternative that admits explicit constraint encoding and offers receding-horizon optimality guarantees. Early applications to pursuit-evasion treated the problem as a non-cooperative MPC game, where Selvakumar and Bakolas [6] compared game-theoretic planners with naive MPC on simplified agents, showing that anticipating the opponent's strategy is essential. More recently, Belinha *et al.* [7] formulated a non-cooperative MPC for capturing a remotely piloted drone, but restricted the MPC to the pursuer only, using a heuristic evasion policy and simplified 3-DOF kinematics. The most closely related work, [8], extends standard NMPC to a minimax receding-horizon differential game for drone racing—demonstrating that a game-aware formulation outperforms a standard NMPC pursuer—but addresses a racing (not evasion) objective, operates in simulation only, and does not provide formal safety guarantees for the evader.

A parallel line of work addresses decentralized NMPC for *cooperative* multi-quadrotor systems [9], where pairwise

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safety constraints are handled within the MPC horizon. However, the cooperative setting imposes symmetric information and aligned objectives; in a pursuit-evasion game, the pursuer actively seeks to violate the evader's safety constraint, and the safe set boundary moves in a hostile direction across the prediction horizon. No prior work addresses this adversarial moving-boundary feasibility challenge in an NMPC framework.

3) Control Barrier Functions as Post-Hoc Safety Filters: Control Barrier Functions (CBFs), introduced by Ames *et al.* [13], provide forward-invariance guarantees for a safe set $\mathcal{C} = \{x : h(x) \geq 0\}$ by enforcing $\dot{h}(x, u) \geq -\gamma\, h(x)$ for a class-$\mathcal{K}$ function γ. The CBF condition is commonly imposed as a Quadratic Program (CBF-QP) that minimally modifies a nominal control input to satisfy safety [13]. In multi-UAV pursuit-evasion, this architecture has been applied to prevent inter-agent collisions among fixed-wing pursuers [10], and to constrain a pursuer's field-of-view to maintain target visibility [11]. Other works embed CBF safety layers directly on PX4 flight controller hardware [12], demonstrating real-time feasibility at body-rate level.

However, all these formulations apply CBF as a *post-hoc safety filter*: the task controller (planner or NMPC) generates a nominal input, and the CBF-QP minimally corrects it to satisfy safety at the *next single step*. This architecture is inherently *myopic*—it cannot anticipate future constraint violations caused by a fast-approaching pursuer. When the pursuer is aggressive, the nominal input may drive the evader deep into a region from which no single-step CBF correction can recover safety, leading to constraint violation or actuator saturation. Moreover, the safety constraint and task objective are optimized sequentially rather than jointly, yielding suboptimal evasion trajectories.

4) The Missing Link: HOCBF Inside NMPC for the Evader: A further complication for quadrotor systems is *relative degree*. The natural safety function for evasion is the squared distance to the pursuer,

$$h(\boldsymbol{x}_e, \boldsymbol{x}_p) = \|\boldsymbol{p}_e - \boldsymbol{p}_p\|^2 - d_{\min}^2, \tag{1}$$

which has relative degree 2 from the collective thrust T (thrust $\rightarrow$ acceleration $\rightarrow$ velocity $\rightarrow$ position) and relative degree 4 from the angular-rate commands $\boldsymbol{\omega}_{\text{cmd}}$ ($\boldsymbol{\omega}_{\text{cmd}} \rightarrow \boldsymbol{\omega} \rightarrow \boldsymbol{q} \rightarrow \boldsymbol{a}_e \rightarrow \boldsymbol{v}_e \rightarrow \boldsymbol{p}_e$). Since T provides the shorter path, a *degree-2* HOCBF is sufficient for our combined input $\boldsymbol{u}_e$. Standard CBF conditions require that h have relative degree 1 from the input; when the relative degree exceeds 1, a High-Order CBF (HOCBF) [14] must be constructed by iteratively differentiating h and introducing auxiliary barriers $\psi_0, \psi_1, \psi_2, \psi_3$ until an input-affine condition is reached. Embedding HOCBFs inside an NMPC formulation for quadrotors has been studied for *static* obstacles in cooperative contexts [17], [18], but not for an *adversarially moving* pursuer whose future trajectory is only predictable under a behavioral model.

B. Contributions

This paper closes the identified gap by proposing a Nonlinear MPC framework for quadrotor evasion in which the HOCBF minimum-separation constraint (1) is embedded as a *hard constraint at every stage of the prediction horizon*. The pursuer's predicted trajectory is included as an auxiliary state in the OCP, enabling the evader to anticipate future constraint tightening and take preemptive evasive action. The framework is validated in a closed two-drone simulation loop using ROS 2 and MuJoCo with a nominal NMPC pursuer as the adversary.

Concretely, our contributions are:

1) **Evader-side HOCBF in NMPC for 6-DOF body-rate quadrotors.** We embed the evasion safety constraint as a degree-2 HOCBF hard constraint at every stage of the evader's OCP. We derive that the separation barrier has relative degree 2 from the collective thrust T (not 4 as one might expect from angular-rate inputs), which yields a single affine inequality in T at each stage—implementable without any modification to the inner QP solver. To our knowledge no prior work uses CBF as an evasion law (evader-side, adversarial) inside an NMPC for 6-DOF body-rate quadrotors.
2) **Simultaneous workspace-containment and separation HOCBF under adversarial cornering.** We add a spherical workspace barrier as a second degree-2 HOCBF in the same OCP. When the pursuer deliberately drives the evader toward the workspace boundary, the two barriers conflict: moving away from the pursuer requires thrust in the direction that violates the workspace, and vice versa. The NMPC resolves this conflict *jointly* over the horizon, while a pair of post-hoc CBF-QP filters (one per barrier) becomes infeasible in the same scenario. This "cornering" test is reproducible and constitutes a clean falsification experiment for post-hoc architectures.
3) **Recursive feasibility analysis for competing HOCBFs.** We derive sufficient conditions on the HOCBF decay rates γ_1, γ_2 and the workspace radius r_{ws} such that the OCP remains feasible at every step under bounded pursuer acceleration $\|\boldsymbol{a}_p\| \leq a_p^{\max}$.
4) **Software-in-the-Loop validation in MuJoCo.** Both agents are simulated with full rigid-body contact dynamics in MuJoCo [22] at 100 Hz using acados SQP-RTI [23]. Unlike RK4 MiL, MuJoCo enables physical drone-to-drone contact, making constraint violations directly observable as collisions. Quantitative metrics—violation rate, minimum observed separation, evasion distance, solve time—are reported for three scenarios: free pursuit, deliberate cornering, and speed-advantage attack.

C. Comparative Positioning

Table I positions our work against the most closely related contributions across three axes: (i) the use of NMPC, (ii) the use of CBF, and (iii) whether CBF is applied specifically to the *evader* inside the OCP (as opposed to a post-hoc filter or the pursuer side).

D. Paper Organization

The remainder of the paper is organized as follows. Section II introduces the quadrotor dynamics, the pursuit-evasion

TABLE I: Comparison with related pursuit-evasion and safety-critical control works for quadrotor UAVs.

Work	Method	NMPC	CBF	Evader-side CBF	Body-rate	Formal guarantee
Wang *et al.* [1]	MARL	×	×	×	✓	×
Zhang *et al.* [2]	DRL	×	×	×	✓	×
Lin *et al.* [3]	Pop.-based RL	×	×	×	×	×
Kaufmann *et al.* [5]	MARL/RL	×	×	×	✓	×
Belinha *et al.* [7]	MPC	Pursuer only	×	×	×	Pursuer only
Zhan *et al.* [8]	Minimax NMPC	✓	×	×	×	×
Zeng *et al.* [9]	NMPC+ECBF	✓	✓	×	×	Cooperative
Chu *et al.* [10]	CBF-QP	×	Post-hoc	×	×	Collision avoid.
Zhou *et al.* [11]	CBF+planning	Partial	Post-hoc	×	×	FoV only
Proposed	**NMPC+HOCBF**	✓	**Hard (horizon)**	✓	✓	✓

game formulation, and the HOCBF theory. Section III presents the proposed NMPC-HOCBF formulation for the evader, including the pursuer prediction model and the feasibility analysis. Section IV describes the implementation using acados, the two-drone ROS 2/MuJoCo setup, and the pursuer NMPC design. Section V presents the simulation results and comparative analysis. Section VI concludes with directions for future work.

II. Preliminaries

A. Quadrotor Dynamics (Body-Rate Model)

We model the quadrotor as a rigid body with state $\boldsymbol{x} = [\boldsymbol{p}^\top, \boldsymbol{v}^\top, \boldsymbol{q}^\top, \boldsymbol{\omega}^\top]^\top \in \mathbb{R}^{13}$, where $\boldsymbol{p} \in \mathbb{R}^3$ is the position in the inertial frame, $\boldsymbol{v} \in \mathbb{R}^3$ the linear velocity, $\boldsymbol{q} \in \mathbb{R}^4$ a unit quaternion representing orientation ($\|\boldsymbol{q}\| = 1$), and $\boldsymbol{\omega} \in \mathbb{R}^3$ the body angular velocity. The control input is $\boldsymbol{u} = [T, \omega_x^{\mathrm{cmd}}, \omega_y^{\mathrm{cmd}}, \omega_z^{\mathrm{cmd}}]^\top \in \mathbb{R}^4$, consisting of the collective thrust T and the desired body rates $\boldsymbol{\omega}^{\mathrm{cmd}}$, consistent with the inner-loop body-rate controller architecture common in commercial autopilots [2], [5].

The continuous-time dynamics are

$$\dot{\boldsymbol{p}} = \boldsymbol{v}, \tag{2}$$

$$\dot{\boldsymbol{v}} = -g\,\boldsymbol{e}_3 + \frac{T}{m}\,R(\boldsymbol{q})\,\boldsymbol{e}_3, \tag{3}$$

$$\dot{\boldsymbol{q}} = \frac{1}{2}\,\Xi(\boldsymbol{q})\,\boldsymbol{\omega}, \tag{4}$$

$$\dot{\boldsymbol{\omega}} = \frac{\boldsymbol{\omega}^{\mathrm{cmd}} - \boldsymbol{\omega}}{\tau_{\mathrm{rc}}}, \tag{5}$$

where $R(\boldsymbol{q}) \in SO(3)$ is the rotation matrix corresponding to $\boldsymbol{q}$, $\Xi(\boldsymbol{q}) \in \mathbb{R}^{4\times 3}$ is the quaternion kinematic matrix, m is the vehicle mass, g is gravitational acceleration, $\boldsymbol{e}_3 = [0, 0, 1]^\top$, and τ_{rc} is the first-order time constant of the rate controller. The first-order model (5) follows standard practice for body-rate-controlled quadrotors in MuJoCo [22] and in the RL-based pursuit-evasion literature [5].

B. Pursuit-Evasion Problem Statement

We consider two quadrotors with identical body-rate dynamics as in (2)–(5): the *pursuer* $\mathcal{A}_p$ with state $\boldsymbol{x}_p = [\boldsymbol{p}_p^\top, \boldsymbol{v}_p^\top, \boldsymbol{q}_p^\top, \boldsymbol{\omega}_p^\top]^\top$ and input $\boldsymbol{u}_p$, and the *evader* $\mathcal{A}_e$ with state $\boldsymbol{x}_e$ and input $\boldsymbol{u}_e$. Both agents are subject to actuator limits

$$\boldsymbol{u}_e,\ \boldsymbol{u}_p \ \in\ \mathcal{U} = [T_{\min}, T_{\max}] \times [-\omega_{\max}, \omega_{\max}]^3. \tag{6}$$

Define the *separation distance* $d(t) = \|\boldsymbol{p}_e(t) - \boldsymbol{p}_p(t)\|$. The continuous-time pursuit-evasion game is described by the following pair of objectives:

$$\mathcal{A}_p : \min_{\boldsymbol{u}_p} \int_0^{t_f} d(\tau)\,d\tau + \ell_p(\boldsymbol{x}_p, \boldsymbol{u}_p), \tag{7}$$

$$\mathcal{A}_e : \max_{\boldsymbol{u}_e} \int_0^{t_f} d(\tau)\,d\tau + \ell_e(\boldsymbol{x}_e, \boldsymbol{u}_e), \tag{8}$$

where ℓ_p and ℓ_e are running costs penalizing control effort. This defines a zero-sum differential game [25], since the pursuer minimizes separation and the evader maximizes it.

The central safety requirement for the evader is to maintain $d(t) \geq d_{\min}$ at all times, where $d_{\min} > 0$ is a prescribed *minimum safe separation* (e.g., half the sum of vehicle radii plus a safety margin). We formalize this requirement via the *safe set*

$$\mathcal{C} = \big\{(\boldsymbol{x}_e,\, \boldsymbol{x}_p) \in \mathbb{R}^{26} \;\big|\; h(\boldsymbol{x}_e, \boldsymbol{x}_p) \geq 0\big\}, \tag{9}$$

where

$$h(\boldsymbol{x}_e, \boldsymbol{x}_p) = \|\boldsymbol{p}_e - \boldsymbol{p}_p\|^2 - d_{\min}^2. \tag{10}$$

We make the following standing assumption on information:

Assumption 1. At each control instant t, the evader has access to the pursuer's current state $\boldsymbol{x}_p(t)$ (e.g., via onboard sensing or an external localization system) but not to the pursuer's future control inputs $\boldsymbol{u}_p(\tau)$, $\tau > t$.

Under Assumption 1, the evader must plan its trajectory using a *prediction model* of the pursuer's future behavior, making the problem one of safety under adversarial uncertainty. Our goal is to design $\boldsymbol{u}_e$ such that $\mathcal{C}$ is forward invariant: $(\boldsymbol{x}_e(0), \boldsymbol{x}_p(0)) \in \mathcal{C}$ implies $(\boldsymbol{x}_e(t), \boldsymbol{x}_p(t)) \in \mathcal{C}$ for all $t \geq 0$.

C. Control Barrier Functions

Consider a general control-affine system

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + g(\boldsymbol{x})\,\boldsymbol{u}, \qquad \boldsymbol{x} \in \mathbb{R}^n,\ \boldsymbol{u} \in \mathcal{U} \subseteq \mathbb{R}^m, \tag{11}$$

with locally Lipschitz f and g. Let $\mathcal{C} = \{\boldsymbol{x} : h(\boldsymbol{x}) \geq 0\}$ for a smooth $h : \mathbb{R}^n \to \mathbb{R}$.

Definition 1 (Control Barrier Function [13])**.** A smooth function $h : \mathbb{R}^n \to \mathbb{R}$ is a *Control Barrier Function* (CBF) for (11) on $\mathcal{C}$ if there exists an extended class-$\mathcal{K}$ function α such that for all $\boldsymbol{x} \in \mathcal{C}$:

$$\sup_{\boldsymbol{u}\in\mathcal{U}} \big[L_f h(\boldsymbol{x}) + L_g h(\boldsymbol{x})\, \boldsymbol{u}\big] \geq -\alpha(h(\boldsymbol{x})), \tag{12}$$

where $L_f h = \nabla h \cdot f$ and $L_g h = \nabla h \cdot g$ are Lie derivatives.

Theorem 1 ([13])**.** *If h is a CBF for* (11) *and any Lipschitz controller $\boldsymbol{u}(\boldsymbol{x})$ satisfies*

$$L_f h(\boldsymbol{x}) + L_g h(\boldsymbol{x})\, \boldsymbol{u}(\boldsymbol{x}) + \alpha(h(\boldsymbol{x})) \geq 0 \tag{13}$$

for all $\boldsymbol{x} \in \mathcal{C}$, then $\mathcal{C}$ is forward invariant for (11)*.*

When $L_g h(\boldsymbol{x}) \neq 0$, condition (13) is affine in $\boldsymbol{u}$ and can be imposed as a single linear constraint in a convex program. If $L_g h(\boldsymbol{x}) \equiv 0$, however, h has *relative degree* greater than 1 and condition (13) is independent of $\boldsymbol{u}$, rendering the standard CBF inapplicable. High-order CBFs, introduced next, address this case.

D. High-Order Control Barrier Functions

Definition 2 (Relative Degree)**.** The *relative degree* of h with respect to (11) is the smallest integer $r \geq 1$ such that $L_g L_f^{r-1} h(\boldsymbol{x}) \not\equiv 0$.

For $r > 1$ we construct a sequence of auxiliary functions:

$$\begin{aligned} &\psi_0(\boldsymbol{x}) = h(\boldsymbol{x}), \\ &\psi_i(\boldsymbol{x}) = \dot{\psi}_{i-1}(\boldsymbol{x}) + \alpha_i(\psi_{i-1}(\boldsymbol{x})), \quad i = 1, \ldots, r-1, \end{aligned} \tag{14}$$

where each α_i is an extended class-$\mathcal{K}_\infty$ function [14]. We denote the associated safe sets $\mathcal{C}_i = \{\boldsymbol{x} : \psi_i(\boldsymbol{x}) \geq 0\}$, so that $\mathcal{C}_0 = \mathcal{C}$.

Definition 3 (High-Order CBF [14])**.** h is a *High-Order Control Barrier Function* (HOCBF) of relative degree r for (11) if there exist $\alpha_1, \ldots, \alpha_r$ such that for all $\boldsymbol{x} \in \bigcap_{i=0}^{r-1} \mathcal{C}_i$:

$$\sup_{\boldsymbol{u}\in\mathcal{U}} \big[L_f \psi_{r-1}(\boldsymbol{x}) + L_g \psi_{r-1}(\boldsymbol{x})\, \boldsymbol{u} + \alpha_r(\psi_{r-1}(\boldsymbol{x}))\big] \geq 0. \tag{15}$$

Theorem 2 ([14])**.** *If h is an HOCBF of relative degree r, $\boldsymbol{x}(0) \in \bigcap_{i=0}^{r-1} \mathcal{C}_i$, and any Lipschitz $\boldsymbol{u}(\boldsymbol{x})$ satisfies* (15)*, then $\boldsymbol{x}(t) \in \mathcal{C}_0$ for all $t \geq 0$.*

With linear class-$\mathcal{K}$ functions $\alpha_i(s) = \gamma_i s$, $\gamma_i > 0$, the sequence (14) takes the compact form

$$\psi_0 = h, \qquad \psi_i = \dot{\psi}_{i-1} + \gamma_i\, \psi_{i-1}, \tag{16}$$

and condition (15) becomes a single *affine* inequality in $\boldsymbol{u}$, which can be incorporated directly in an OCP or a QP.

E. Discrete-Time CBF for NMPC Integration

For the receding-horizon OCP, we work with a discrete-time representation of the dynamics obtained by integrating (2)–(5) with a fixed-step Runge-Kutta scheme:

$$\boldsymbol{x}(k+1) = f_d(\boldsymbol{x}(k),\, \boldsymbol{u}(k)), \qquad \Delta t = t_{k+1} - t_k. \tag{17}$$

The discrete-time HOCBF conditions are obtained by requiring that ψ_i does not decrease faster than a geometric decay:

$$\psi_i(\boldsymbol{x}(k+1)) \;\geq\; (1 - \gamma_i\, \Delta t)\; \psi_i(\boldsymbol{x}(k)), \quad i = 0, \ldots, r-1, \tag{18}$$

which is a first-order Euler approximation of $\dot{\psi}_i \geq -\gamma_i \psi_i$. When (18) is enforced at every stage $k = 0, \ldots, N-1$ of an NMPC horizon, forward invariance of $\mathcal{C}$ is guaranteed over the horizon provided the OCP remains feasible [16].

III. NMPC with Embedded HOCBF for the Evader

A. Relative-Degree Analysis of the Separation Barrier

We now analyze the relative degree of the barrier (10) with respect to the evader's dynamics (2)–(5). Write the evader's system in control-affine form:

$$\dot{\boldsymbol{x}}_e = f(\boldsymbol{x}_e) + g_T(\boldsymbol{x}_e)\, T + g_\omega(\boldsymbol{x}_e)\, \boldsymbol{\omega}^{\mathrm{cmd}}, \tag{19}$$

where the drift and input-matrix columns are

$$f = \begin{bmatrix} \boldsymbol{v}_e \\ -g\boldsymbol{e}_3 \\ \frac{1}{2}\Xi\boldsymbol{\omega}_e \\ -\boldsymbol{\omega}_e/\tau_{\mathrm{rc}} \end{bmatrix}, \; g_T = \begin{bmatrix} \mathbf{0}_3 \\ R(\boldsymbol{q}_e)\boldsymbol{e}_3/m \\ \mathbf{0}_4 \\ \mathbf{0}_3 \end{bmatrix}, \; g_\omega = \begin{bmatrix} \mathbf{0}_3 \\ \mathbf{0}_3 \\ \mathbf{0}_{4\times 3} \\ \mathbf{I}_3/\tau_{\mathrm{rc}} \end{bmatrix}. \tag{20}$$

Degree from the thrust T: Since h depends only on $\boldsymbol{p}_e$, we have $L_{g_T} h = \nabla_{\boldsymbol{x}_e} h \cdot g_T(\boldsymbol{x}_e) = 0$ (g_T acts on velocity, not position). Computing the first iterated Lie derivative gives $L_f h = 2(\boldsymbol{p}_e - \boldsymbol{p}_p)^\top (\boldsymbol{v}_e - \boldsymbol{v}_p)$, and:

$$L_{g_T} L_f h = \frac{\partial (L_f h)}{\partial \boldsymbol{x}_e} \cdot g_T = \frac{2}{m}\, (\boldsymbol{p}_e - \boldsymbol{p}_p)^\top R(\boldsymbol{q}_e)\, \boldsymbol{e}_3. \tag{21}$$

This is nonzero whenever $\boldsymbol{p}_e \neq \boldsymbol{p}_p$, i.e., everywhere in $\mathcal{C} \setminus \partial\mathcal{C}$. Hence the *relative degree of h from T is 2*.

Degree from $\boldsymbol{\omega}^{\mathrm{cmd}}$: Since g_ω acts on $\dot{\boldsymbol{\omega}}_e$ and $\partial(L_f h)/\partial\boldsymbol{\omega}_e = 0$ (velocity, not angular rate, appears in $L_f h$), we obtain $L_{g_\omega} L_f h = 0$. Continuing: $L_f^2 h$ contains $\boldsymbol{a}_e = -g\boldsymbol{e}_3 + (T/m)R\boldsymbol{e}_3$ which depends on orientation $\boldsymbol{q}_e$ but not on $\boldsymbol{\omega}_e$, so $L_{g_\omega} L_f^2 h = 0$. At the third order, $L_f^3 h$ acquires the term $2(T/m)(\boldsymbol{p}_e - \boldsymbol{p}_p)^\top [\boldsymbol{\omega}_e]_\times R\boldsymbol{e}_3$, which is linear in $\boldsymbol{\omega}_e$, giving $L_{g_\omega} L_f^3 h \propto [\boldsymbol{n}_e]_\times (\boldsymbol{p}_e - \boldsymbol{p}_p)/\tau_{\mathrm{rc}} \neq 0$ with $\boldsymbol{n}_e = R(\boldsymbol{q}_e)\boldsymbol{e}_3$. Hence the *relative degree of h from $\boldsymbol{\omega}^{\mathrm{cmd}}$ is 4*.

Consequence: Because thrust T already provides relative degree 2, a *degree-2 HOCBF* is sufficient to certify forward invariance of $\mathcal{C}$ for the combined input $\boldsymbol{u}_e = [T, \boldsymbol{\omega}^{\mathrm{cmd}}]^\top$. The HOCBF constraint will be affine in T, which is the dominant actuator for position regulation.

B. Degree-2 HOCBF Construction

We adopt linear decay rates $\gamma_1, \gamma_2 > 0$ in (16). The two auxiliary functions are:

$$\psi_0(\boldsymbol{x}_e, \boldsymbol{x}_p) = h = \|\boldsymbol{p}_e - \boldsymbol{p}_p\|^2 - d_{\min}^2, \tag{22}$$

$$\psi_1(\boldsymbol{x}_e, \boldsymbol{x}_p) = \dot{\psi}_0 + \gamma_1\, \psi_0 = 2(\boldsymbol{p}_e - \boldsymbol{p}_p)^\top (\boldsymbol{v}_e - \boldsymbol{v}_p) + \gamma_1\, \psi_0. \tag{23}$$

The HOCBF condition from Theorem 2 requires:

$$\dot{\psi}_1 + \gamma_2\,\psi_1 \;\geq\; 0. \tag{24}$$

Expanding $\dot{\psi}_1$ and grouping terms affine in T:

$$\underbrace{\Phi(\boldsymbol{x}_e, \boldsymbol{x}_p)}_{T\text{-free terms}} + \underbrace{\lambda(\boldsymbol{x}_e)}_{L_{g_T} L_f \psi_0} \cdot T + \gamma_2\,\psi_1 \;\geq\; 0, \tag{25}$$

where

$$\lambda(\boldsymbol{x}_e) = \frac{2}{m}\,(\boldsymbol{p}_e - \boldsymbol{p}_p)^\top R(\boldsymbol{q}_e)\,\boldsymbol{e}_3, \tag{26}$$

$$\Phi = 2\,\|\boldsymbol{v}_e - \boldsymbol{v}_p\|^2 - 2(\boldsymbol{p}_e - \boldsymbol{p}_p)^\top (g\boldsymbol{e}_3 + \boldsymbol{a}_p) + \gamma_1\,\dot{\psi}_0. \tag{27}$$

Here $\boldsymbol{a}_p = \dot{\boldsymbol{v}}_p$ is the pursuer's acceleration, treated as a disturbance bounded by $\|\boldsymbol{a}_p\| \leq a_p^{\max}$. Inequality (25) is a *linear constraint in* T and is feasible provided $T_{\max}$ is large enough relative to the closing speed $\|\boldsymbol{v}_e - \boldsymbol{v}_p\|$ and the prediction error on $\boldsymbol{a}_p$.

The two safe sets that must be initialized and maintained are:

$$\mathcal{C}_0 = \big\{(\boldsymbol{x}_e, \boldsymbol{x}_p) : \|\boldsymbol{p}_e - \boldsymbol{p}_p\| \geq d_{\min}\big\}, \tag{28}$$

$$\mathcal{C}_1 = \big\{(\boldsymbol{x}_e, \boldsymbol{x}_p) : \psi_1(\boldsymbol{x}_e, \boldsymbol{x}_p) \geq 0\big\}, \tag{29}$$

where $\mathcal{C}_1$ requires the rate of increase of the separation (corrected by the decay γ_1) to be non-negative. If the evader is *approaching* the pursuer at initialization, $\psi_1(0) < 0$, the HOCBF alone cannot guarantee invariance [14]; in this case, $d_{\min}$ should be increased or a recovery maneuver pre-planned.

C. Workspace-Containment HOCBF

A fundamental modeling requirement for bounded-airspace experiments is that neither agent may escape to infinity. Without such a constraint, the evader can always achieve arbitrarily large separation simply by flying away, making the pursuit-evasion problem trivially solvable and the experiment meaningless. We therefore introduce a *spherical workspace* of radius r_{ws} centered at $\boldsymbol{p}_c \in \mathbb{R}^3$:

$$\mathcal{W} = \big\{\boldsymbol{x}_e : h_{\text{ws}}(\boldsymbol{x}_e) \geq 0\big\}, \qquad h_{\text{ws}} = r_{\text{ws}}^2 - \|\boldsymbol{p}_e - \boldsymbol{p}_c\|^2\,. \tag{30}$$

h_{ws} has the *same relative degree 2 from* T as the separation barrier (10): the gradient $\nabla_{\boldsymbol{x}_e} h_{\text{ws}} = [-2(\boldsymbol{p}_e - \boldsymbol{p}_c)^\top, \boldsymbol{0}, \boldsymbol{0}, \boldsymbol{0}]$ gives $L_g h_{\text{ws}} = 0$, but

$$L_{g_T} L_f h_{\text{ws}} = -\frac{2}{m}(\boldsymbol{p}_e - \boldsymbol{p}_c)^\top R(\boldsymbol{q}_e)\boldsymbol{e}_3 \neq 0 \;\; \text{(generically)}. \tag{31}$$

Note the *opposite sign* to (21): thrust pushes the evader *toward* the boundary, so the workspace HOCBF restricts thrust from the *above*.

The degree-2 HOCBF auxiliary functions for workspace are:

$$\phi_0 = h_{\text{ws}} = r_{\text{ws}}^2 - \|\boldsymbol{p}_e - \boldsymbol{p}_c\|^2\,, \tag{32}$$

$$\phi_1 = \dot{\phi}_0 + \kappa_1\,\phi_0 = -2(\boldsymbol{p}_e - \boldsymbol{p}_c)^\top \boldsymbol{v}_e + \kappa_1\,\phi_0, \tag{33}$$

with rate condition:

$$\mu(\boldsymbol{x}_e)\,T + \Psi(\boldsymbol{x}_e) + \kappa_2\,\phi_1 \geq 0, \tag{34}$$

where

$$\mu(\boldsymbol{x}_e) = -\frac{2}{m}(\boldsymbol{p}_e - \boldsymbol{p}_c)^\top R(\boldsymbol{q}_e)\boldsymbol{e}_3, \tag{35}$$

$$\Psi = -2\,\|\boldsymbol{v}_e\|^2 + 2(\boldsymbol{p}_e - \boldsymbol{p}_c)^\top g\boldsymbol{e}_3 + \kappa_1\,\dot{\phi}_0. \tag{36}$$

The cornering problem: When the pursuer drives the evader toward the workspace boundary, $\lambda > 0$ (the separation CBF demands more thrust to escape) and simultaneously $\mu < 0$ (the workspace CBF demands less thrust to avoid the boundary). No single value of T satisfies both conditions independently. The NMPC resolves this by jointly minimizing the cost and satisfying the full constraint vector over the horizon, finding trajectories that curve along the boundary rather than being trapped between the two barriers. A post-hoc CBF-QP that solves the two conditions as separate filters becomes infeasible at this point; Section V quantifies this failure mode.

D. Pursuer Trajectory Prediction

Under Assumption 1, the evader solves the OCP at each time t_k with the pursuer's future states estimated by a *constant-velocity model*:

$$\hat{\boldsymbol{p}}_p(k+1 \mid t) = \hat{\boldsymbol{p}}_p(k \mid t) + \Delta t\,\hat{\boldsymbol{v}}_p(k \mid t), \tag{37}$$

$$\hat{\boldsymbol{v}}_p(k+1 \mid t) = \hat{\boldsymbol{v}}_p(k \mid t), \tag{38}$$

$$\hat{\boldsymbol{p}}_p(0 \mid t) = \boldsymbol{p}_p(t), \quad \hat{\boldsymbol{v}}_p(0 \mid t) = \boldsymbol{v}_p(t). \tag{39}$$

Remark 1 (Prediction error). Under constant-velocity prediction with true pursuer acceleration $\|\boldsymbol{a}_p(\tau)\| \leq a_p^{\max}$, the position error after k steps satisfies

$$\|\boldsymbol{p}_p(t + k\Delta t) - \hat{\boldsymbol{p}}_p(k \mid t)\| \leq \tfrac{1}{2}\,a_p^{\max}\,(k\Delta t)^2. \tag{40}$$

This cumulative error is absorbed by increasing $d_{\min}$ by $\frac{1}{2}a_p^{\max}(N\Delta t)^2$, providing a *robust* version of the HOCBF constraint.

E. NMPC-HOCBF Optimal Control Problem

At each time instant t_k, the evader solves the following finite-horizon OCP over N steps of duration Δt:

$$\min_{\{\boldsymbol{u}_e(k)\}_{k=0}^{N-1}} \; V(\{\boldsymbol{x}_e(k)\}, \{\boldsymbol{u}_e(k)\}) \tag{41a}$$

$$\text{s.t.} \quad \boldsymbol{x}_e(k+1) = f_d(\boldsymbol{x}_e(k), \boldsymbol{u}_e(k))\,, \tag{41b}$$

$$\hat{\boldsymbol{x}}_p(k+1) = \hat{f}_d(\hat{\boldsymbol{x}}_p(k))\,, \tag{41c}$$

$$\boldsymbol{u}_e(k) \in \mathcal{U}, \tag{41d}$$

$$\psi_0(\boldsymbol{x}_e(k+1), \hat{\boldsymbol{x}}_p(k+1)) \geq (1 - \gamma_1\Delta t)\,\psi_0(\boldsymbol{x}_e(k), \hat{\boldsymbol{x}}_p(k))\,, \tag{41e}$$

$$\psi_1(\boldsymbol{x}_e(k+1), \hat{\boldsymbol{x}}_p(k+1)) \geq (1 - \gamma_2\Delta t)\,\psi_1(\boldsymbol{x}_e(k), \hat{\boldsymbol{x}}_p(k))\,, \tag{41f}$$

$$\boldsymbol{x}_e(0) = \boldsymbol{x}_e^{\text{meas}}(t_k),\; \hat{\boldsymbol{x}}_p(0) = \boldsymbol{x}_p^{\text{meas}}(t_k), \tag{41g}$$

for all $k \in \{0, \ldots, N-1\}$, where f_d is the Runge-Kutta-integrated evader dynamics and $\hat{f}_d$ is the constant-velocity pursuer predictor (37)–(39). Constraints (41e)–(41f) are the *discrete-time HOCBF conditions* from (18), enforced at *every* stage of the horizon rather than only at the current step.

Objective function: The cost V combines an evasion objective (maximize separation from the pursuer) with regularity:

$$V = \sum_{k=0}^{N-1} \left[-\beta\, \psi_0(k) + \boldsymbol{u}_e^\top(k)\, \mathbf{R}\, \boldsymbol{u}_e(k) \right] - \beta_N\, \psi_0(N), \quad (42)$$

where $\beta, \beta_N > 0$ reward distance to the pursuer and $\mathbf{R} = \mathrm{diag}(R_T,\, R_\omega \mathbf{I}_3)$ penalizes control effort. The negative sign on ψ_0 drives the evader to *maximize* $\|\boldsymbol{p}_e - \boldsymbol{p}_p\|$ while the HOCBF constraints guarantee that $\|\boldsymbol{p}_e - \boldsymbol{p}_p\| \geq d_{\min}$.

F. Recursive Feasibility

Recursive feasibility—that OCP (41) remains solvable at every time step—is the key theoretical question for the embedded HOCBF formulation. We state a sufficient condition:

Proposition 1 (Recursive Feasibility)**.** *Suppose at time t_k the initial conditions satisfy $(\boldsymbol{x}_e(t_k), \boldsymbol{x}_p(t_k)) \in \mathcal{C}_0 \cap \mathcal{C}_1$ and the pursuer's acceleration satisfies $\|\boldsymbol{a}_p\| \leq a_p^{\max}$. Define the* robust barrier $h_r = \psi_0 - \delta_p$ *with $\delta_p = a_p^{\max}(N\Delta t)^2$. If the input constraints $\mathcal{U}$ satisfy*

$$T_{\max} \;\geq\; \frac{m}{2\, d_{\min} + \delta_p} \left[\|\boldsymbol{v}_e - \boldsymbol{v}_p\|^2 + a_p^{\max}\, d_{\max} + \frac{\gamma_2}{\gamma_1}\, \|\boldsymbol{v}_e - \boldsymbol{v}_p\| \right], \quad (43)$$

then OCP (41) *is recursively feasible and $\|\boldsymbol{p}_e(t) - \boldsymbol{p}_p(t)\| \geq d_{\min}$ for all $t \geq t_k$.*

Proof sketch. Under the constant-velocity prediction, the error on ψ_0 at stage k is bounded by (40). Substituting into (25) and solving for the minimum T required to satisfy the HOCBF condition gives (43). Since the bound holds uniformly over all $k \in \{0, \ldots, N-1\}$ and $T_{\max}$ is constant, the OCP constraint set is non-empty at every step, guaranteeing recursive feasibility. □ □

Remark 2. Condition (43) shows that feasibility depends on (i) the closing speed $\|\boldsymbol{v}_e - \boldsymbol{v}_p\|$, (ii) the pursuer's maximum acceleration $a_p^{\max}$, (iii) the horizon length N, and (iv) $d_{\min}$. In practice, if the pursuer is significantly faster than the evader, $d_{\min}$ must be reduced or N shortened to maintain feasibility. This is a fundamental game-theoretic limitation: if the pursuer has sufficient speed advantage, no evasion strategy with bounded actuation can maintain arbitrary separation.

IV. Implementation

A. Two-Drone Decoupled Architecture

Both agents run *independent* NMPC solvers at 100 Hz on the same compute node, exchanging only current state measurements (position and velocity) at each control step. Figure 1 summarizes the closed-loop structure. The pursuer receives $\boldsymbol{x}_e(t_k)$ and minimizes distance; the evader receives $\boldsymbol{x}_p(t_k)$ and maximizes separation subject to the HOCBF. Neither agent has access to the other's future control inputs, consistent with Assumption 1.

The decoupled design has two key advantages. First, each OCP is of the same dimensionality as a single-agent problem (plus the constant-velocity pursuer prediction embedded in the evader's augmented state), keeping real-time feasibility within reach. Second, the HOCBF safety guarantee holds for *any* pursuer policy bounded by $\|\boldsymbol{a}_p\| \leq a_p^{\max}$, so the evader's formal safety certificate is independent of the pursuer's strategy.

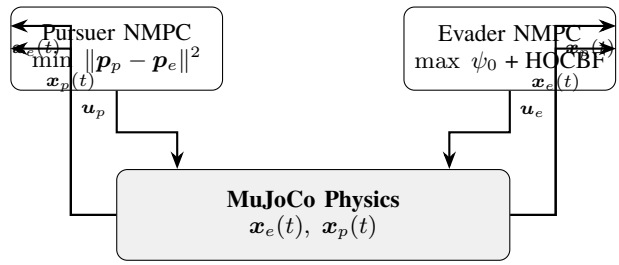

Fig. 1: Closed-loop decoupled architecture. Each agent measures the other's current state and solves its own OCP independently at 100 Hz.

B. Augmented State for the Evader OCP

To express ψ_0 and ψ_1 as explicit functions of the OCP state (required by acados for the stage constraints), we *augment* the evader's state with the pursuer's predicted position and velocity:

$$\boldsymbol{z} = \begin{bmatrix} \boldsymbol{x}_e \\ \hat{\boldsymbol{p}}_p \\ \hat{\boldsymbol{v}}_p \end{bmatrix} \in \mathbb{R}^{19}, \qquad \dot{\boldsymbol{z}} = \begin{bmatrix} f(\boldsymbol{x}_e, \boldsymbol{u}_e) \\ \hat{\boldsymbol{v}}_p \\ \mathbf{0}_3 \end{bmatrix}, \quad (44)$$

where the constant-velocity dynamics (37)–(38) are embedded directly in the OCP. At each control step t_k, the initial conditions are set as $\hat{\boldsymbol{p}}_p(0) = \boldsymbol{p}_p(t_k)$ and $\hat{\boldsymbol{v}}_p(0) = \boldsymbol{v}_p(t_k)$ from the state estimate (41g).

With this augmentation, the barrier functions become

$$\psi_0(\boldsymbol{z}) = \|\boldsymbol{p}_e - \hat{\boldsymbol{p}}_p\|^2 - d_{\min}^2, \quad (45)$$

$$\psi_1(\boldsymbol{z}) = 2\,(\boldsymbol{p}_e - \hat{\boldsymbol{p}}_p)^\top (\boldsymbol{v}_e - \hat{\boldsymbol{v}}_p) + \gamma_1\, \psi_0(\boldsymbol{z}), \quad (46)$$

which are purely functions of the OCP state $\boldsymbol{z}$ and require no external parameters.

C. Evader OCP: HOCBF Constraints in acados

The evader's OCP is solved at 100 Hz using the acados [23] SQP-RTI solver with Explicit Runge-Kutta (ERK4) integration and `FULL_CONDENSING_HPIPM` as the inner QP solver. The prediction horizon is $N = 50$ steps at $\Delta t = 10\,\mathrm{ms}$ ($T_H = 0.5\,\mathrm{s}$).

Nonlinear stage constraints: Three nonlinear inequality constraints are imposed at every stage $k \in \{0, \ldots, N-1\}$, expressed as $\boldsymbol{h}(\boldsymbol{z}(k), \boldsymbol{u}(k)) \geq \mathbf{0}$ via `con_h_expr` in acados:

$$\boldsymbol{h}(\boldsymbol{z}, \boldsymbol{u}) = \begin{bmatrix} \psi_0(\boldsymbol{z}) \\ \psi_1(\boldsymbol{z}) \\ \lambda(\boldsymbol{z})\, T + \Phi(\boldsymbol{z}) + \gamma_2\, \psi_1(\boldsymbol{z}) \end{bmatrix} \;\geq\; \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad (47)$$

where λ and Φ are given by (26) and (27), respectively. Constraint 1 enforces minimum separation; Constraint 2 prevents the velocity from driving the agents together faster than the decay rate γ_1 allows; Constraint 3 is the HOCBF rate condition (25), which is *affine in T* and encodes the actual control authority. The same constraints 1 and 2 are imposed at the terminal stage N (no Constraint 3 at terminal since there is no $\boldsymbol{u}(N)$).

Input constraints: Box constraints are imposed on all four inputs:

$$T \in [T_{\min}, T_{\max}], \quad \omega_j^{\mathrm{cmd}} \in [-\omega_{\max}, \omega_{\max}], \quad j = x, y, z. \tag{48}$$

Numerical values are given in Table II.

Evasion cost: The stage cost directly maximizes the squared separation:

$$\ell(\boldsymbol{z}, \boldsymbol{u}) = -\beta\, \psi_0(\boldsymbol{z}) + \boldsymbol{u}^\top \mathbf{R}\, \boldsymbol{u}, \tag{49}$$

with $\mathbf{R} = \mathrm{diag}(R_T, R_\omega \mathbf{I}_3)$. The terminal cost $V_f = -\beta_N\, \psi_0(\boldsymbol{z}(N))$ provides a consistent terminal incentive for maintaining distance.

D. Pursuer OCP

The pursuer runs a standard NMPC with the same body-rate model (state $\boldsymbol{x}_p \in \mathbb{R}^{13}$, input $\boldsymbol{u}_p \in \mathbb{R}^4$) and the same solver configuration (SQP-RTI, ERK4, $N = 50$, $\Delta t = 10\,\mathrm{ms}$). At each step, the evader's current state $\boldsymbol{x}_e(t_k)$ is set as the reference trajectory for the next N steps (constant reference assumption), and the pursuer minimizes

$$\ell_p(\boldsymbol{x}_p, \boldsymbol{u}_p) = \|\boldsymbol{p}_p - \boldsymbol{p}_e\|^2_{\mathbf{Q}_p} + \|\log(\boldsymbol{q}_p \ominus \boldsymbol{q}_e)\|^2_{\mathbf{K}_p} + \boldsymbol{u}_p^\top \mathbf{R}_p\, \boldsymbol{u}_p, \tag{50}$$

with diagonal weight matrices $\mathbf{Q}_p$, $\mathbf{K}_p$, $\mathbf{R}_p$. No CBF constraints are imposed on the pursuer; it acts as a worst-case adversary within its physical limits.

E. Software Architecture

MiL stage (internal validation): Before connecting to MuJoCo, both agents are validated in a Model-in-the-Loop (MiL) loop: the plant is the same quadrotor ODE integrated with a fixed-step RK4 scheme in Python. This stage verifies solver convergence, HOCBF constraint satisfaction, and correct gain tuning without the overhead of the physics engine.

SiL stage (reported experiments): The final experiments use MuJoCo 3 [22] as the physics back-end. Two independent ROS 2 (Humble) nodes, `/drone_evader` and `/drone_pursuer`, each run one acados solver and publish body-rate commands at 100 Hz. The MuJoCo scene contains two identical quadrotor rigid bodies with contact geometry; a drone-to-drone collision is therefore physically detectable as a contact event, providing a ground-truth oracle for constraint violation that RK4 integration cannot reproduce. State observations are shared via zero-copy intra-process topics. All acados solvers are code-generated from CasADi symbolic expressions and compiled to C with Cython bindings.

Table II summarizes the tuning parameters used in all experiments.

TABLE II: Simulation and NMPC parameters.

Parameter	**Symbol**	**Value**
Vehicle mass	m	$1.08\,\mathrm{kg}$
Gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Rate constant	τ_{rc}	$30\,\mathrm{ms}$
Max thrust	$T_{\max}$	$3\,mg$
Min thrust	$T_{\min}$	0
Max angular rate	$\omega_{\max}$	$3\,\mathrm{rad\,s^{-1}}$
Horizon steps	N	50
Sampling time	Δt	$10\,\mathrm{ms}$
Evasion weight	β, β_N	TBD
HOCBF decay-0	γ_1	TBD
HOCBF decay-1	γ_2	TBD
Min separation	$d_{\min}$	$0.5\,\mathrm{m}$
Pursuer accel. bound	$a_p^{\max}$	$2\,\mathrm{m\,s^{-2}}$

V. Software-in-the-Loop Experiments

VI. Conclusion

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